pfi

Python for Fixed Income

COURSE 1 · PYTHON FOUNDATIONS

Start Here: Python, Notebooks, and Libraries

WEEK 1 · DAY 0

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

The morning note

From the desk lead

Before the 9:00 meeting, tell me what a 5.25 percent coupon pays each half year, and send me a picture of our three bonds.

1. A number

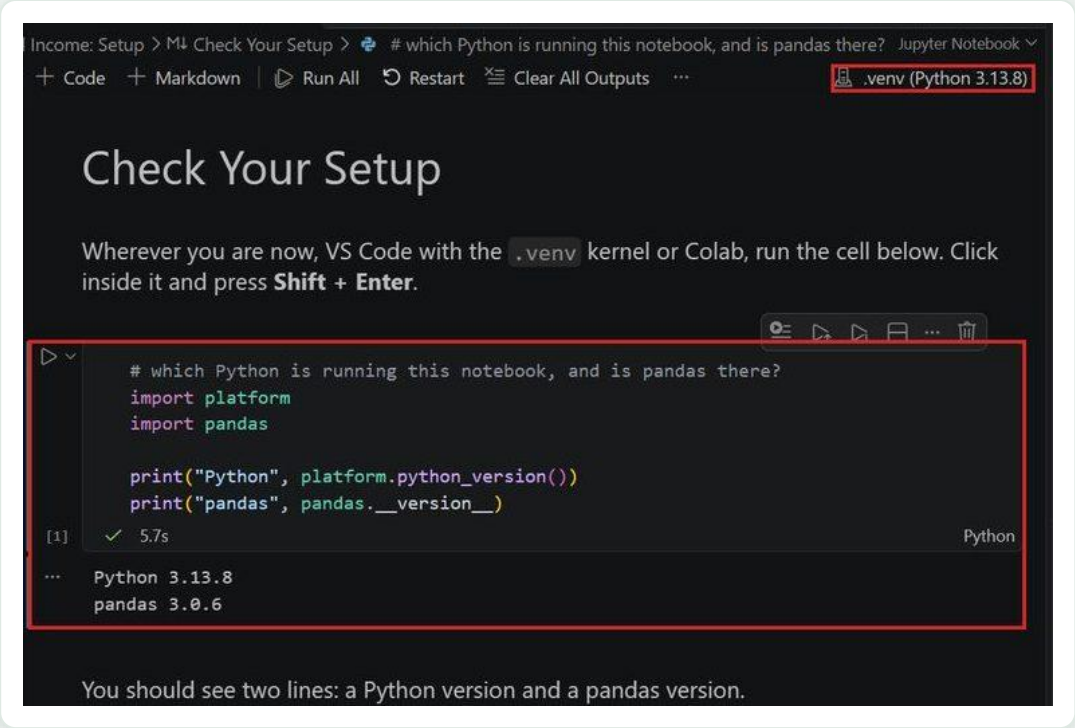
What a 5.25 percent coupon pays each half year

2. A picture

A chart of the desk's three bonds

You have never written a line of code. By the end of this notebook, Python will have answered both.

0.1 Your first cell



Click in a cell, press Shift + Enter

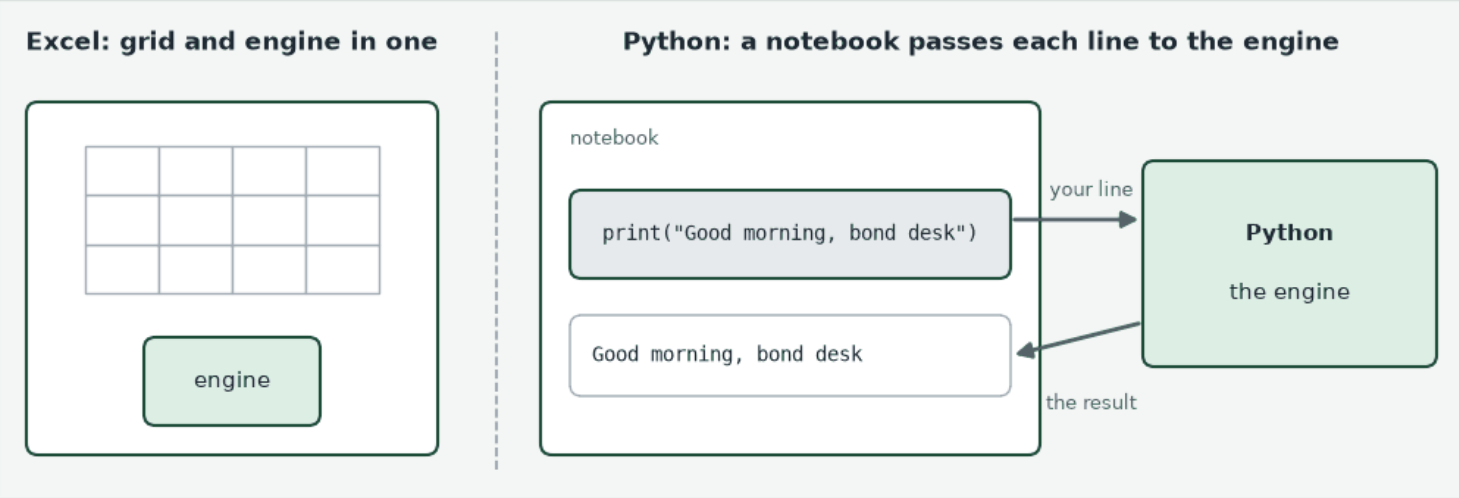
```
# a line that starts with # is a comment
print("Good morning, bond desk")
Good morning, bond desk
```



Every cell, from the top: Run All. Colab: Runtime, then Run all, or Ctrl + F9. VS Code: Run All at the top.

Something read that line and carried it out. That something is Python.

0.2 A language and an engine



Excel side by side

Excel is the grid and the engine in one application. Python is only the engine.

Why "Python"?

Started over the 1989 Christmas holidays, posted in February 1991, and named for Monty Python's Flying Circus, not the snake.

Sources: Python FAQ, "History of the software", read 25 October 2026.

0.3 Python as a calculator

Python	Answer	Excel
`2 + 2`	4	`=2+2`
`5 - 3`	2	`=5-3`
`5 * 5`	25	`=5*5`
`10 / 5`	2.0	`=10/5`
`10 // 5`	2	`=QUOTIENT(10,5)`
`7 // 2`	3	`=QUOTIENT(7,2)`
`7 % 2`	1	`=MOD(7,2)`
`5 ** 2`	25	`=5^2`

The ^ trap

```
# a power typed the Excel way
5 ^ 2
7
```

7, not 25, and no error: a wrong answer that looks right.

For a power, always type **

Three habits, then the first sum

Brackets first

```
2 + 3 * 4
14
(2 + 3) * 4
20
```

When in doubt, add brackets.

No commas in numbers

```
1,000,000
(1, 0, 0)
1_000_000
1000000
```

Python reads straight past underscores: they are only for your eyes.

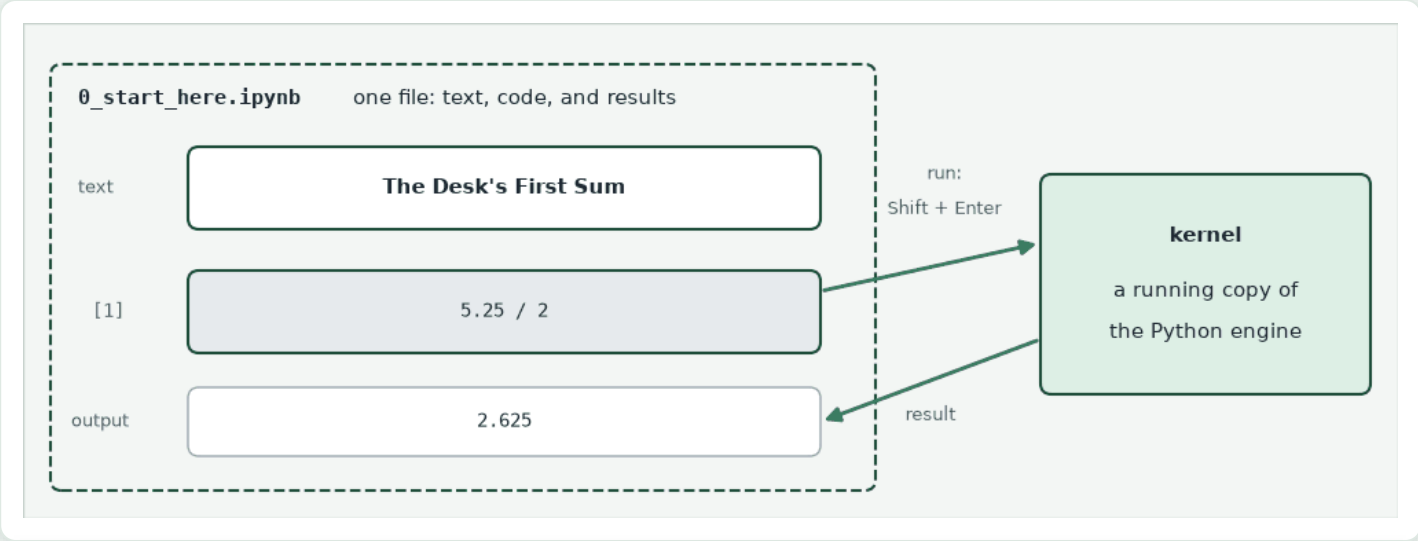
Half the coupon

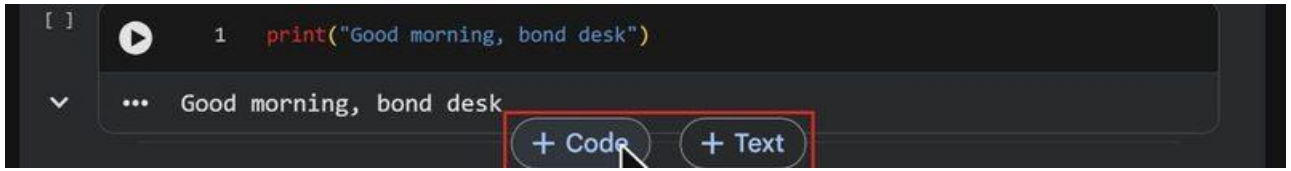
```
5.25 / 2
2.625
```

2.625 percent of face each half year.

Half the note answered. Next: the page you have been running.

0.4 Text, code, and results in one file





Rest the mouse between two cells, and two buttons appear.

A cell remembers

```
coupon = 5.25
coupon / 2
2.625
```

The kernel remembers in the order you ran the cells, not their order on the page.

Confused? Restart, then Run All.

0.5 A file of code only

The file, half_coupon.py

```
coupon = 5.25
print("Half the coupon:", coupon / 2)
```

Typed in a terminal

```
python half_coupon.py
```

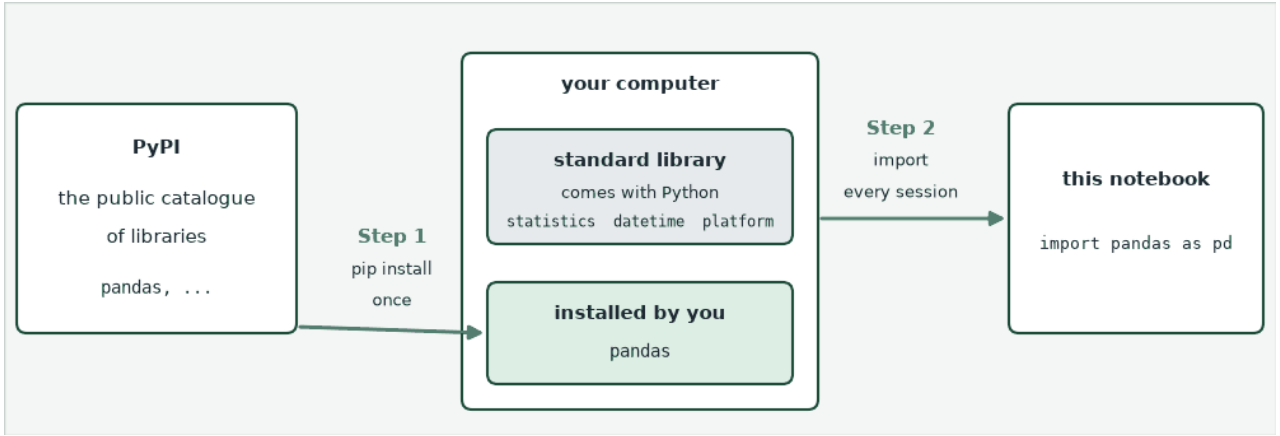
What the terminal shows

```
Half the coupon: 2.625
```

	Notebook (.ipynb)	Python file (.py)
Holds	text, code, and results	code only
Runs	one cell at a time, when you choose	the whole file, top to bottom
Good for	working step by step	code you keep and reuse

Python on its own is small. Libraries do the rest.

0.6 Install once, import every time



Built in

```
prices = [98.50, 101.25, 99.75]
print("Highest:", max(prices))
Highest: 101.25
```

statistics

```
import statistics
statistics.mean(prices)
99.83333333333333
```

datetime

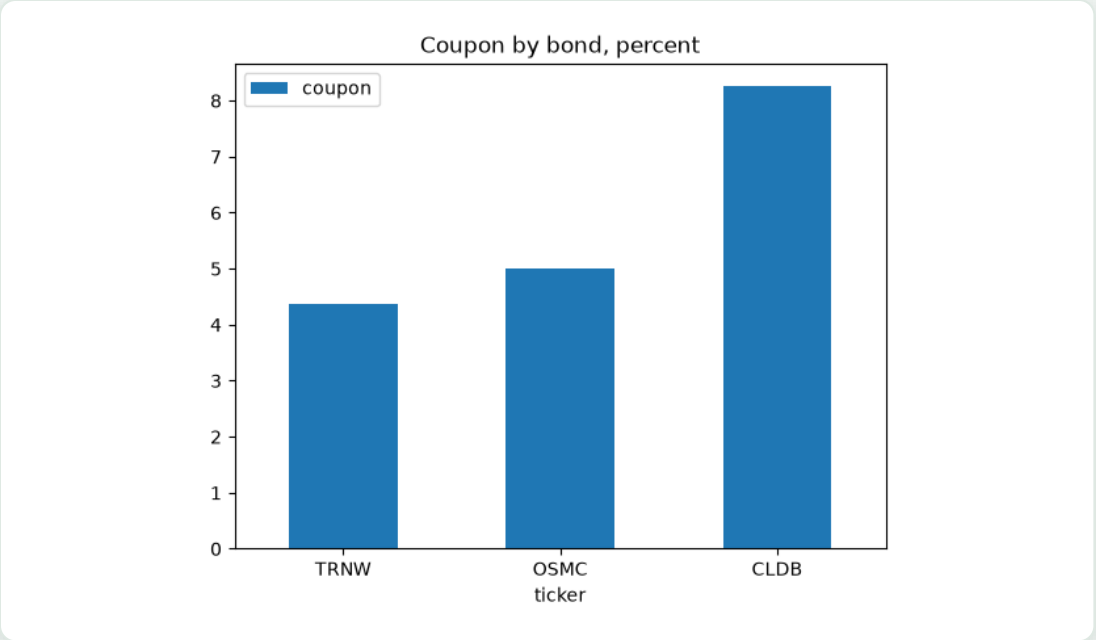
```
from datetime import date
days = date(2026, 7, 15) - date(2026, 1, 15)
print(days.days, "days")
181 days
```

Three bonds in a table, then a picture

```
import pandas as pd
bonds = pd.DataFrame({
    "ticker": ["TRNW", "OSMC", "CLDB"],
    "coupon": [4.375, 5.00, 8.25],
    "price": [96.25, 100.00, 104.50],
})
bonds
```

ticker	coupon	price
TRNW	4.375	96.25
OSMC	5.000	100.00
CLDB	8.250	104.50

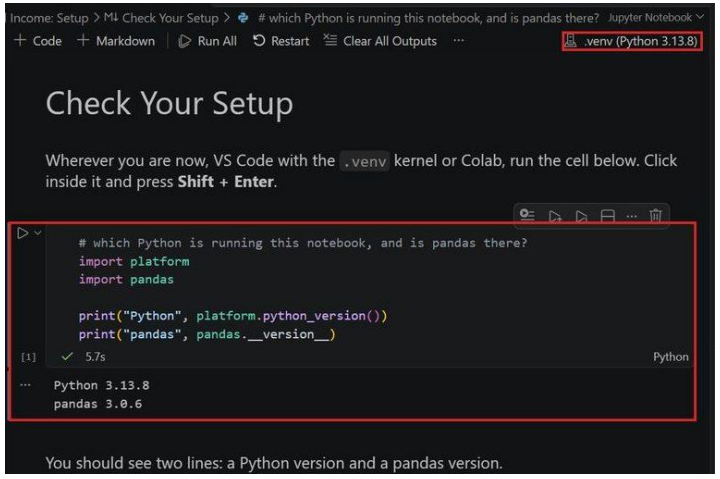
```
import bondtools
ModuleNotFoundError: No module named 'bondtools'
```

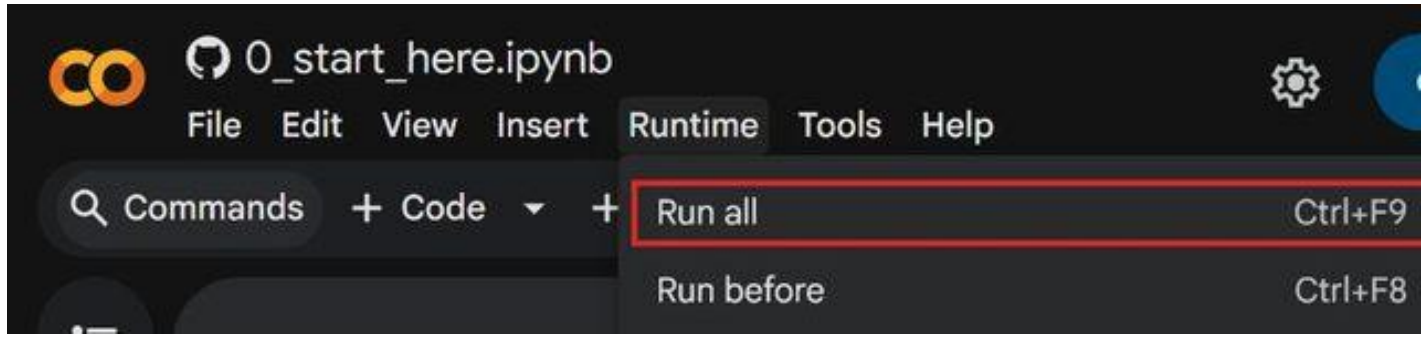



```
chart = bonds.plot.bar(x="ticker", y="coupon",
    rot=0, title="Coupon by bond, percent")
```

0.7 VS Code or Colab

	VS Code	Colab
Runs on	your Windows computer	Google's computers, in a browser
Libraries live in	a .venv folder with its own Python and libraries	Google's computers
Data	on a work computer, your firm's rules come first	public or made-up data only





The Setup notebook walks through both, step by step.

0.8 The day in Excel terms

Excel	Python
A workbook (.xlsx)	A notebook (.ipynb)
Excel, the application that calculates	Python, the engine
A formula, =5.25/2	A code cell, 5.25 / 2
A text box on the sheet	A markdown cell
Pressing F9	Run All
SUM, AVERAGE	sum, statistics.mean
An add-in, such as the Analysis ToolPak	A library
A VBA module	A .py file

One big difference: Excel, on its default automatic setting, recalculates by itself; a Python cell runs only when you run it.

What you can do now

- Run a cell, or every cell with Run All
- Say what Python is: a language and its engine
- Use Python as a calculator, with ** for a power
- Tell a notebook from a .py file
- Install and import a library, and draw a chart

Day 1 You meet the whole desk: four bonds, and your first real calculations on them.